

Solution sketches for the exam

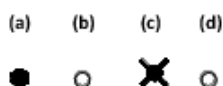
August 30, 2025

General Exam Instructions

- Please check carefully whether your exam sheets are complete and correct; objections after the exam cannot be considered.
- There are different exam **versions**. The different versions are marked with numbers. The number at the bottom left identifies your exam version.
- The **question sheet** (including the pages with the general remarks) consists of 8 pages. In addition there is a one-page **solution sheet** and you get concept paper for calculations.
- It is allowed to use a non-programmable calculator without internet access and **two** DIN-A4 pages with individual notes (i.e., one sheet of paper written double-sided or two sheets of paper written one-sided). The use of other resources (e.g. programmable calculators or calculators with internet access, personal concept paper) constitutes an attempt at deception and will be reported to the examination board.
- You have 120 minutes to solve the exam.
- The exam consists of 30 questions. On the problem set, you find 5 answer possibilities for each exam question. For each question, **exactly one** of the given answer possibilities is correct. Here, “correct” means that the statement is true in each case that is covered by the underlying model that was part of the course curriculum.
- You have to fill in your solutions in the corresponding solution fields on the solution sheet.

Handling of the Solution Sheet

- You **only** have to hand in the solution sheet at the end of the exam. Answers on concept sheets or on the question sheet will not be corrected. We recommend that you fill in the solutions towards the **end of the exam** in order to avoid corrections. Please start to fill in your answers **at least 10 minutes before the end of the exam**. The supervisors have orders to collect the solution sheets, even if you have not yet filled in your answers.
- *Please fill in the whole circle, do not mark answers with a cross! Only unambiguously legible solutions can yield points.* Please do not use TippEx to correct your answers! In the case that a circle was already filled in, but then you want to give no answer, first fill in another circle and then cross out both circles (see example). Please use dark colors (black or blue) and no pencil.
- Example: The answer is supposed to be (a), but answer (c) was filled in. Then in the end the solution sheet has to look like this:



- You must **sign** your **solution sheet** at the bottom.

Grading

- In total you can obtain up to *30 points*.
- For each correct answer, you obtain *one point*.
- If you obtain at least *27 points*, you get the grade 1.0 in any case.
- You will pass the exam with certainty if you obtain at least the fraction *14/30 of the available points* or if you are among the 80% best participants who showed up for the exam (relative to the total number of participants of the German and English exam combined).
- If it is necessary to exclude a problem from the assessment due to an error or an ambiguity in the question, then the maximum number of points available, the number of points necessary for the grade 1.0, and the minimum number of points required to pass the exam will be reduced by 1 point.

Good Luck!

1. Consider a quasi-linear economy with free entry. All firms produce with the same production function, which has the unique efficient scale $\check{q} = 5$. There are 100 consumers, each of whom has the individual demand function $D^i(p) = 1/p$. If exactly 10 firms enter the market in equilibrium, then the equilibrium price is

- (a) 5.
- (b) 1.
- (c) 3.
- (d) 4.
- (e) 2.

Since 10 firms enter the market, the equilibrium quantity is $q^* = 10\check{q} = 50$. The market demand is $D(p) = 100/p$. The equilibrium price p^* is such that $D(p^*) = q^*$. Therefore, $100/p^* = 50$, so $p^* = 2$.

Correct answers: (e)

2. The interest rate $r = 1\%$ corresponds to the money discount factor

- (a) 1.
- (b) $100/101$.
- (c) $101/100$.
- (d) $11/10$.
- (e) $10/11$.

The interest rate $r = 1/100$ corresponds to the money discount factor $1/(1+r) = 1/(101/100) = 100/101$.

Correct answers: (b)

3. Consider the conditional factor demand $d(p_1, p_2, q)$ of a firm with a strictly increasing and quasi-concave production function. Then the following applies: if p_1 is increased while p_2 and q remain unchanged, then

- (a) it cannot be optimal to increase the quantity of input 1 used.
- (b) none of the other answers are correct.
- (c) the iso-cost line becomes flatter.
- (d) at least the same quantity of each input is used as before.
- (e) the firm's costs decrease.

Law of demand for conditional factor demand.

Correct answers: (a)

4. A consumer's preferences regarding consumption streams (c_0, c_1) in the periods 0 and 1 are represented by the utility function $u(c_0, c_1) = \max\{c_0, c_1\}$. The consumer has an income stream with a strictly positive net present value and can save and borrow money at the interest rate $r > 0$. Then the following holds:

- (a) The consumer will spend twice as much money in period 0 relative to period 1.
- (b) The consumer will spend half as much money in period 0 relative to period 1.
- (c) The consumer will not spend any money in period 1.
- (d) The consumer will not spend any money in period 0.
- (e) The consumer will spend the same amount of money in period 0 as in period 1.

Due to the max preferences, the indifference curves are inverted L-shaped, so that it is always optimal to consume either only in period 0 or only in period 1. However, since consumption in period 1 costs less than in period 0 due to the interest rate, the consumer will only consume in period 1.

Correct answers: (d)

5. Consider a quasi-linear economy with three consumers. Among other vectors, the following utility vectors can be achieved through allocations: $(0, 10, 20)$, $(1, 9, 19)$, $(5, 5, 20)$, $(9, 1, 20)$, $(10, 10, 10)$. Then the following utility vector cannot be achieved by an efficient allocation:

- (a) $(1, 9, 19)$
- (b) $(9, 1, 20)$
- (c) $(5, 5, 20)$
- (d) $(0, 10, 20)$
- (e) $(10, 10, 10)$

The allocation that leads to $(0, 10, 20)$ can be changed so that individuals 2 and 3 each give half a monetary unit to individual 1. This results in the utility vector $(1, 9.5, 19.5)$. This Pareto-dominates $(1, 9, 19)$, which therefore belongs to an inefficient allocation. All other vectors mentioned have the same total utility of 30. We cannot exclude the possibility that these vectors are Pareto efficient.

Correct answers: (a)

6. In a quasi-linear economy, consider an individual i with the willingness-to-pay function $v^i(x^i) = 1 - e^{-x^i}$ for all $x^i \geq 0$, where e^z (defined for any real number z) is the exponential function (i.e., the derivative with respect to z is again e^z). The individual consumes $x^i = 2$ units of good 1, which the individual purchased at the market price $\hat{p} = e^{-3}$ (per unit). The individual has a money endowment of $m_i = 3$ and has no initial endowment of good 1. The individual does not own any shares of any firm. The individual then receives the (money-equivalent) utility

- (a) $3 - 2e^{-2} - 4e^{-3}$.
- (b) $2 - 3e^{-2} - 4e^{-3}$.
- (c) $2 - e^{-2} - e^{-3}$.
- (d) $4 - e^{-2} - 2e^{-3}$.
- (e) $3 - 4e^{-2} - 2e^{-3}$.

The consumer surplus is $CS^i = v^i(x^i) - \hat{p}x^i = 1 - e^{-2} - 2e^{-3}$. The utility

Correct answers: (d)

7. The preference relation of a consumer for a bundle of goods (x_1, x_2) is quasi-linear, where x_2 represents the money spent on the goods other than good 1. The consumer has the marginal willingness to pay function $v'(x_1) = 1/(x_1 + 10)$. Assume $v(0) = 0$. Then the consumer has the willingness-to-pay function $v(x_1) =$

- (a) $\ln(x_1 + 10) - \ln(10)$.
- (b) x_1 .
- (c) $(x_1 + 10)^2 - 100$.
- (d) $1/(x_1 + 10)^2$.
- (e) $-1/(x_1 + 10)^2$.

The derivative of $v(x_1) = \ln(x_1 + 10) - \ln(10)$ is correct, and $v(0) = 0$.

Correct answers: (a)

8. Consider an Edgeworth exchange economy. Each trader has the initial endowment $(1, 1)$. The A -traders have preferences represented by the utility function $u^A(x_1, x_2) = \min\{x_1, x_2\}$. The B -traders have preferences represented by the utility function $u^B(x_1, x_2) = \min\{x_1, 2x_2\}$. Then the contract curve consists of the line segment connecting the following points (described from the perspective of the coordinate origin of the B -traders):

- (a) $(0, 0)$ and $(2, 2)$
- (b) $(1, 1)$ and $(1, 1/2)$
- (c) $(1, 1)$ and $(3/2, 1)$
- (d) $(1, 1)$ and $(1, 3/2)$
- (e) $(1, 1)$ and $(1/2, 1)$

A sketch of the indifference curves through the endowment point makes this clear. The A -indifference curve is L-shaped with a kink at $(1, 1)$. The B -indifference curve is L-shaped with a kink at $(1, 1/2)$.

Correct answers: (b)

9. A consumer has the utility function $u(x_1, x_2) = \min\{x_1, 7x_2\}$ over bundles of goods (x_1, x_2) with $x_i \geq 0$ for $i = 1, 2$. Then the demand for good 1 is given by $d_1(p_1, p_2, Y) =$

- (a) $Y/(p_1 + p_2)$.
- (b) $Y/(7p_1 + p_2)$.
- (c) $7Y/(7p_1 + p_2)$.
- (d) $7Y/(p_1 + 7p_2)$.
- (e) $Y/(p_1 + 7p_2)$.

For the optimal bundle, $x_1^* = 7x_2^*$ and $p_1x_1^* + p_2x_2^* = Y$. Therefore $(7p_1 + p_2)x_2^* = Y$. So $x_2^* = Y/(7p_1 + p_2)$ and therefore $x_1^* = 7Y/(7p_1 + p_2)$.

Correct answers: (c)

10. Consider a firm that has zero setup costs. The firm's marginal cost function is $MC(q) = (1/q)^{1/3}$ for all $q < 111$. The firm can offer up to 111 units of output, but cannot offer more than 111 units of output (capacity limit). Then, at the output price $p = 1/3$, the quantity supplied is equal to
- (a) 9.
 - (b) 31.
 - (c) 1.
 - (d) 111.
 - (e) 27.

It is important to note that MC is decreasing. The efficient scale is therefore at $\tilde{q} = 111$ and price does not equal MC .

Correct answers: (d)

11. A firm has the production function $f(x_1, x_2) = \sqrt{x_1} + (x_2)^2$. Then the firm has
- (a) no strictly increasing production function.
 - (b) everywhere decreasing returns to scale.
 - (c) neither everywhere increasing, nor everywhere constant, nor everywhere decreasing returns to scale.
 - (d) everywhere increasing returns to scale.
 - (e) everywhere constant returns to scale.

Consider $t > 1$. In the range in which x_1 is large and x_2 is small, $f(tx_1, tx_2) \approx \sqrt{tx_1} < t\sqrt{x_1} \approx tf(x_1, x_2)$, i.e. $f(tx_1, tx_2) < tf(x_1, x_2)$. Conversely, $f(tx_1, tx_2) > tf(x_1, x_2)$ applies in the range in which x_1 is small and x_2 is large.

Correct answers: (c)

12. Consider a consumer with a monotonic preference relation over bundles of goods (x_1, x_2) with $x_i \geq 0$ for $i = 1, 2$. Assume that good 1 is a normal good. Denote the consumer's demand as d . Then the following holds:

- (a) $d_1(4, p_2, Y) \geq d_1(3, p_2, Y)$ for all $p_2 > 0$ and $Y > 0$.
- (b) $d_2(4, p_2, Y) \leq d_2(3, p_2, Y)$ for all $p_2 > 0$ and $Y > 0$.
- (c) $d_2(4, p_2, Y) = d_2(3, p_2, Y)$ for all $p_2 > 0$ and $Y > 0$.
- (d) None of the other answers is correct.
- (e) $d_1(4, p_2, Y) \leq d_1(3, p_2, Y)$ for all $p_2 > 0$ and $Y > 0$.

Because good 1 is normal, the law of demand applies to good 1, i.e. to d_1 .

Correct answers: (e)

13. Consider a quasi-linear economy in which there are exactly two firms. At the equilibrium price p^* , the supply quantity of firm 1 is equal to 7 and the supply quantity of firm 2 is equal to 9. Furthermore, at the equilibrium price p^* , the output price elasticity of supply of firm 1 is equal to 2 and the output price elasticity of supply of firm 2 is equal to 4. Then, at the equilibrium price p^* , the elasticity of the market supply with respect to the output price is

- (a) 50.
- (b) $25/8$.
- (c) 3.
- (d) 6.
- (e) a number different from those given in the other answers.

If S^1 , S^2 , η^1 and η^2 denote the individual supplies and price elasticities, then the following applies to the price elasticity of the market supply

$$\eta(p^*) = \frac{S^1(p^*)}{S^1(p^*) + S^2(p^*)} \eta^1(p^*) + \frac{S^2(p^*)}{S^1(p^*) + S^2(p^*)} \eta^2(p^*) = \frac{7}{16} 2 + \frac{9}{16} 4 = \frac{25}{8}.$$

Correct answers: (b)

14. A consumer has the utility function $u(x_1, x_2) = (x_1)^2$ over two-goods bundles (x_1, x_2) with $x_1 \geq 0$ and $x_2 \geq 0$. Which statement is *not* correct?

- (a) The consumer's preference relation is convex.
- (b) The consumer's preference relation is monotonic.
- (c) Good 1 is a normal good everywhere.
- (d) Good 2 is a normal good everywhere.
- (e) The consumer will always consume a strictly positive quantity of both goods.

The consumer will only consume good 1.

Correct answers: (e)

15. Consider the individual demand of a consumer with monotonic preferences over bundles of two goods. If the price of a good changes in such a way that the consumer becomes effectively richer, then the following relationship applies between the consumer's budget Y and the consumer's Hicks-compensated budget Y^H :

- (a) $Y^H < Y$
- (b) $Y^H = Y$
- (c) $Y^H > Y$
- (d) $Y^H \leq Y$
- (e) $Y^H \geq Y$

The Hicks compensation neutralizes the effective increase in wealth, i.e. $Y^H \leq Y$. However, $Y^H < Y$ does not necessarily apply. If, for example, the price of good 1 is reduced and the consumer does not consume any of good 1 before and after the price change, then there is no need to compensate, i.e. $Y^H = Y$.

Correct answers: (d)

16. A consumer with Cobb-Douglas preferences has the marginal rate of substitution $MRS_{1,2}(1,7) = -1$ at the point $(1,7)$. Then, at the point $(7,1)$, the consumer has the marginal rate of substitution $MRS_{1,2}(7,1) =$

- (a) -49 .
- (b) $-1/7$.
- (c) -7 .
- (d) -1 .
- (e) $-1/49$.

$MRS_{1,2}(x_1, x_2) = -(c/d)(x_2/x_1)$. So $-(c/d)7 = -1$. So $c/d = 1/7$. So $MRS_{1,2}(7,1) = -(1/7)(1/7) = -1/49$.

Correct answers: (e)

17. A consumer has a convex preference relation over bundles of goods (x_1, x_2) (with $x_i \geq 0$ for $i = 1, 2$). The following utility function u then surely does *not* represent the consumer's preference relation:

- (a) $u(x_1, x_2) = (x_1)^2$
- (b) $u(x_1, x_2) = -(x_2)^2$
- (c) $u(x_1, x_2) = -(x_1 - 5)^2 + x_2$
- (d) $u(x_1, x_2) = (x_1)^2 - x_2$
- (e) $u(x_1, x_2) = x_1 - x_2$

In the example $u(x_1, x_2) = x_1 - x_2$, the indifference curves are parallel to the main diagonal in the (x_1, x_2) -diagram and the upper-contour sets are to the right of it, i.e. convex. In the example $u(x_1, x_2) = (x_1)^2$, the indifference curves are vertical and the upper-contour sets to the right of them, i.e. convex. In the example $u(x_1, x_2) = -(x_2)^2$, the indifference curves are horizontal and the upper-contour sets are downwards relative to them, i.e. convex. In the example $u(x_1, x_2) = -(x_1 - 5)^2 + x_2$, the upper-contour sets are enclosed by parabolas (and coordinate axes), i.e. they are convex. In the example $u(x_1, x_2) = (x_1)^2 - x_2$ the upper-contour sets are each to the right of the right branch of a parabola, i.e. not convex.

Correct answers: (d)

18. A consumer has quasi-linear preferences. The derivative of her willingness-to-pay function is strictly decreasing everywhere. Then the following holds:

- (a) The demand function is strictly increasing.
- (b) None of the other answers apply.
- (c) The demand function is constant.
- (d) The demand function is strictly decreasing in the area in which strictly positive quantities of the good are demanded.
- (e) The demand function is strictly decreasing everywhere.

At all prices $p > v'(0)$ the consumer demands the quantity 0, implying that the demand function is not strictly decreasing there.

Correct answers: (d)

19. Consider an Edgeworth exchange economy in which all traders (i.e., the A -traders and the B -traders) have the same monotonic and convex preferences over their respective bundles of goods and all traders have the initial endowment (that is, $u^A = u^B$ and $e^A = e^B$). Then the following holds:

- (a) There is an equilibrium in which all traders consume their initial endowments.
- (b) None of the statements in the other answers is correct.
- (c) There is an equilibrium in which the A traders consume nothing of good 2 and the B traders consume nothing of good 1.
- (d) There is an equilibrium in which each trader consumes the same quantity of good 1 as of good 2.
- (e) There is no equilibrium.

The equilibrium price ratio equals the absolute value of the marginal rate of substitution at the initial-endowment point.

Correct answers: (a)

20. A typical barrier to free entry is

- (a) a widely known production technology.
- (b) a high price.
- (c) a large number of competitors.
- (d) a patent.
- (e) none of the options listed in the other answers.

Correct answers: (d)

21. If a firm has strictly positive setup costs, then it can happen that

- (a) the marginal cost is equal to 1 everywhere and the firm offers strictly positive output quantities at arbitrarily low output prices.
- (b) the cost function has no jump point at 0.
- (c) the average-cost function is decreasing everywhere.
- (d) the average-cost function is constant.
- (e) the average-cost function is increasing everywhere.

See lecture chapters 7 and 8.

Correct answers: (c)

22. You observe that a consumer finds the bundle (10,10) strictly worse than the following two bundles: (12,9) and (3,12). Then her preference relation cannot have the following two properties at the same time:

- (a) convexity and completeness
- (b) monotonicity and transitivity
- (c) monotonicity and completeness
- (d) convexity and monotonicity
- (e) convexity and transitivity

We consider the convex combination $\frac{2}{3}(12,9) + \frac{1}{3}(3,12) = 2(4,3) + (1,4) = (9,10)$. If the pref-rel. is convex, then it will find the bundle (9,10) at least as good as the worse of the two bundles (12,9) and (4,3). If the pref-rel. is monotonic, then it will find the bundle (10,10) at least as good as the bundle (9,10). So convexity and monotonicity cannot be fulfilled together. However, each of the two properties monotonicity and convexity individually could be fulfilled. (And then, by definition of these properties, rationality is also fulfilled, i.e. also completeness and transitivity).

Correct answers: (d)

23. There exist exchange economies in which at least one Pareto-efficient allocation

- (a) makes every trader at least as well off as in the initial allocation and still does not belong to the contract curve.
- (b) is Pareto-inefficient.
- (c) can be Pareto-improved.
- (d) cannot be achieved in equilibrium by redistributing the initial endowments.
- (e) is Pareto-dominated.

In the lecture we have seen a graphical example of an Edgeworth exchange economy with non-convex preferences in which this happened.

Correct answers: (d)

24. A firm can use three input goods for production. It has a strictly increasing production function f with increasing returns to scale. Then the following condition is satisfied:

- (a) $f(2,3,4) < f(6,5,11)/2$
- (b) $f(2,3,4) > f(6,9,13)/3$
- (c) $f(2,3,4) > f(6,5,11)/2$
- (d) $f(2,3,4) = f(6,5,11)/2$
- (e) $f(2,3,4) < f(6,9,13)/3$

Due to the scale returns, $3f(2,3,4) < f(6,9,12)$. Due to the strictly increasing production function, $f(6,9,12) < f(6,9,13)$. The example $f(x_1, x_2, x_3) = (x_1)^2/100 + x_2 + x_3/100$ shows that $f(2,3,4) < f(6,5,11)/2$ does not necessarily hold. The statements with ">" or "=" are obviously false.

Correct answers: (e)

25. You observe a consumer with the utility function $u(x_1, x_2) = \min\{x_1, \sqrt{x_2}\}$ over bundles of goods (x_1, x_2) with $x_i \geq 0$ for $i = 1, 2$. You observe that she consumes 4 units of good 1. The prices of the goods are $p_1 = 2$ and $p_2 = 1$. Then the consumer has the following budget:

- (a) 12
- (b) a budget different from those stated in the other answers
- (c) 16
- (d) 24
- (e) 20

She has L-shaped indifference curves with kinks along the curve $x_1 = \sqrt{x_2}$. For the optimal bundle (x_1^*, x_2^*) , $x_1^* = \sqrt{x_2^*}$ applies, i.e. $x_1^* = 4$ and $x_2^* = 16$. Walras' law states that $2x_1^* + x_2^* = Y$. Therefore, $24 = Y$.

Correct answers: (d)

26. At a firm's efficient scale, the following holds:

- (a) marginal costs are maximal.
- (b) the costs are minimal.
- (c) marginal costs are minimal.
- (d) average costs are maximal.
- (e) average costs are minimal.

See lecture chapter 6.

Correct answers: (e)

27. Consider a quasi-linear economy. The government intervenes by setting a market price for all transactions that is strictly above the equilibrium price. What effect of this intervention cannot occur, relative to the competitive equilibrium without government intervention?

- (a) Some consumers remain equally well off.
- (b) Total welfare is increased.
- (c) Some firms are worse off.
- (d) Some consumers are worse off.
- (e) Some firms are better off.

Chapter 10B. Firms may be worse off because they may sell less (albeit at a higher price). If there are consumers who did not buy anything in the competitive equilibrium without government intervention, then they remain equally well off.

Correct answers: (b)

28. Consider a demand curve $D(p)$ that is unitary elastic at price $p = 10$. Then $D(10) - D(11) \approx$

- (a) $D(10)/100$.
- (b) $D(10)/20$.
- (c) $D(10)/200$.
- (d) $D(10)$.
- (e) $D(10)/10$.

The change to price $p' = 11$ is a 10% price increase and, due to unit elasticity, leads to an approximate 10% decrease in the quantity demanded, i.e. $(D(11) - D(10))/D(10) \approx -1/10$.

Correct answers: (e)

29. In the intertemporal choice model with two periods, 0 and 1, consider a consumer with the utility function $u(c_0, c_1) = c_0 + \ln(c_1 + 1)$. At the current interest rate, the consumer is a borrower. Which of the following statements is certainly true regarding the effect of an interest rate increase (in a comparative-statics sense)?
- (a) The consumer will become a saver.
 - (b) The consumer will not strictly increase the amount of her loans.
 - (c) None of the statements given in the other answers is correct.
 - (d) The consumer will be made strictly better off.
 - (e) The consumer will remain equally well off.

The utility function is additively separable. Thus, both goods are normal and the considerations described at the end of Chapter 4 in the lecture apply.

Correct answers: (b)

30. Consider the conditional factor demand of a firm that has a strictly increasing production function. The elasticity of the demand for input 1 with respect to the price of input 2 is
- (a) ≥ 1 .
 - (b) ≤ -1 .
 - (c) ≤ 0 .
 - (d) $= 0$.
 - (e) ≥ 0 .

Due to the law of demand for conditional factor demand, an increase in the price of input 2 will cause the quantity demanded of input 2 to remain the same or decrease, and therefore the quantity demanded of input 1 will remain the same or increase.

Correct answers: (e)